A Global Hamiltonian Period Atlas for Positive Lotka–Volterra Flow

HCS Research Program

28 August 2026 (revision 1)

Abstract

We give a global, source-native description of the positive two-species Lotka–Volterra flow at its physical time clock. A logarithmic normalization turns the system into a strictly convex proper Hamiltonian flow. Every positive energy is therefore one compact periodic oval. The two real Lambert-W branches of the exponential potential give exact one-dimensional quadratures for the enclosed area and period, and coarea gives the action identity $J'(h) = T(h)/(2\pi)$. Linearization supplies the center-period limit, while integrating the log equations gives exact prey and predator cycle averages. An independent certificate checks 24 levels in six rational parameter sets and re-integrates the period with a SciPy event ODE. We do not assert period monotonicity or a high-energy asymptotic, and we make no target arithmetic or operator claim.

1 Model and global claim

Consider

$$\dot{x} = x(a - by), \qquad \dot{y} = y(-c + dx), \qquad a, b, c, d > 0,$$

on $x, y > 0$, with physical time t. Put $x_* = c/d$, $y_* = a/b$ and $u = \log(x/x_*), v = \log(y/y_*)$. Then

$$\dot{u} = a(1 - e^v), \qquad \dot{v} = c(e^u - 1), \qquad H = c(e^u - u - 1) + a(e^v - v - 1).$$

Theorem 1 (period annulus and exact atlas)**.** *H is a strict convex proper first integral with unique critical point $(0, 0)$. Every level $H = h > 0$ is a smooth compact periodic oval. If $F(s) = e^s - s - 1$, let*

$$\ell(r) = -W_0(-e^{-1-r}) - 1 - r, \qquad q(r) = -W_{-1}(-e^{-1-r}) - 1 - r.$$

With $u_- = \ell(h/c)$, $u_+ = q(h/c)$, and $r(u) = (h - cF(u))/a$, its area and period are

$$A(h) = \int_{u_-}^{u_+} [q(r(u)) - \ell(r(u))]\, du,$$

$$T(h) = \frac{1}{a} \int_{u_-}^{u_+} \left[\frac{1}{1 - e^{\ell(r(u))}} + \frac{1}{e^{q(r(u))} - 1}\right] du.$$

The endpoint singularities are integrable. For $J(h) = A(h)/(2\pi)$, $J'(h) = T(h)/(2\pi)$, and $\lim_{h\downarrow 0} T(h) = 2\pi/\sqrt{ac}$. Every periodic orbit obeys $\langle x\rangle = c/d$ and $\langle y\rangle = a/b$.

Proof. The Hessian is $\mathrm{diag}(ce^u, ae^v)$ and F tends to infinity at both ends, giving strict convexity and properness. The sublevel $K_h = \{H \le h\}$ is a compact convex body with nonempty interior, so its boundary $H = h$ is connected; regularity follows because the only critical point is the origin. The canonical equations are $\dot{u} = -H_v, \dot{v} = H_u$, and the vector field is nonzero on this boundary. Thus the connected compact one-manifold is one periodic oval. Solving $F(s) = r$ gives the two displayed Lambert branches. Splitting the oval into lower and upper graphs gives the quadratures. Along the level, $|\dot{(u, v)}| = |\nabla H|$, hence $dt = ds/|\nabla H|$; the coarea formula then gives $dA/dh = \int_{H=h} ds/|\nabla H| = T$. The center matrix is $\left(\begin{smallmatrix} 0 & -a \\ c & 0 \end{smallmatrix}\right)$. Finally, integrating $\dot{u}$ and $\dot{v}$ over a period gives $\langle e^u\rangle = \langle e^v\rangle = 1$. $\square$

2 Branches, action, and boundaries

The branch labels matter: W_0 gives the negative graph and W_{-1} the positive graph. A trigonometric substitution in the certificate cancels the square-root endpoint singularities without changing the physical clock. The geometric action is positive by definition; no orientation convention is needed. The axes are invariant one-dimensional flows, not members of the positive period annulus. If a rate is zero, coercivity or the interior Hamiltonian statement can fail and the resulting boundary equation is treated separately. We intentionally leave period monotonicity and large-energy asymptotics outside the theorem.

Source note

Period-analysis context includes Waldvogel [1, 2] and Hsu [3]. We do not use the historical monotonicity result as a claim here and make no priority statement.

References

[1] J. Waldvogel, "The Period in the Volterra–Lotka Predator-Prey Model," *SIAM Journal on Numerical Analysis* 20(6) (1983), 1264–1272, DOI:10.1137/0720098.

[2] J. Waldvogel, "The period in the Volterra-Lotka system is monotonic," *Journal of Mathematical Analysis and Applications* 114(1) (1986), 178–184, DOI:10.1016/0022-247X(86)90076-4.

[3] S.-B. Hsu, "A Remark on the Period of the Periodic Solution in the Lotka-Volterra System," *Journal of Mathematical Analysis and Applications* 95(2) (1983), 428–436, DOI:10.1016/0022-247X(83)90117-8.

Declarations

Scope. The exact registered literal is `NO_BAD_EULER_OR_ROOT_NUMBER`. **Data and code.** The theorem, ledger, independent checks, and build instructions are released with HCS-C211. **Competing interests.** None declared. **AI-use disclosure.** Generative tools assisted drafting and code generation. The released theorem, numerical values, source metadata, and scope boundaries were checked by the internal artifact chain; this is not external peer review.